

Project Based Learning

On

COMPUTER AIDED DRAFTING SOFTWARE PROJECT

STEPPED FOOTING DESIGN AND DRAFTING USING AUTOCAD

By

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1. ABSTRACT

Stepped footing is a type of shallow foundation commonly used to safely transfer structural loads from columns to the supporting soil while reducing excessive bearing pressure. It consists of a series of concrete layers arranged in a step-like pattern, where each lower step provides a larger bearing area than the one above. Stepped footings are particularly suitable for structures constructed on sloping ground, weak soils, or locations where heavy loads require gradual load distribution. Proper design and detailing of stepped footings contribute to improved structural stability, reduced settlement, and enhanced foundation performance. Based on the sample project format.

This project focuses on the design and drafting of an RCC stepped footing using AutoCAD. The drawing package includes a footing plan, elevation, sectional view, reinforcement detailing, dimensions, and construction notes prepared using standard drafting practices. Various AutoCAD commands such as Line, Rectangle, Offset, Trim, Extend, Hatch, Dimension, and Layer Management are utilized to produce accurate and professional engineering drawings. Through this project, a clear understanding of foundation engineering concepts, reinforcement detailing, load transfer mechanisms, and computer-aided drafting techniques is achieved, demonstrating the importance of AutoCAD in modern civil engineering design and documentation.

2. INTRODUCTION

Stepped footing is a type of isolated foundation used to safely transfer structural loads from columns or walls to the soil through a series of concrete steps. It is commonly adopted when heavy loads are involved or when a gradual increase in footing area is required to distribute the load uniformly and prevent excessive soil pressure. The stepped arrangement improves stability, reduces stress concentration, and provides an economical foundation solution for buildings and other civil engineering structures.

The design of a stepped footing involves determining the footing dimensions, step heights, reinforcement details, and checking the bearing capacity of the soil according to relevant design codes. After completing the structural design calculations, AutoCAD is used to prepare detailed engineering drawings, including plan views, sectional views, reinforcement layouts, dimensions, and annotations. AutoCAD helps engineers create accurate, clear, and professional drafting documents that are essential for construction and site execution. By combining structural design principles with AutoCAD drafting techniques, engineers can effectively communicate footing specifications, ensure design accuracy, and facilitate efficient construction practices.

3. Aim of the Project:

To design and prepare detailed **Stepped Footing drawings using AutoCAD**, including plan, elevation, section, and reinforcement details for effective load transfer from the structure to the supporting soil.

4. OBJECTIVES OF THE PROJECT

- To understand the components and function of a stepped footing.
- To develop accurate footing drawings using AutoCAD tools and commands.
- To prepare plan, elevation, and sectional views of the footing.
- To represent reinforcement details and dimensions clearly.
- To demonstrate the load transfer mechanism from the column to the soil.
- To apply proper drafting standards, layers, annotations, and hatching in AutoCAD.
- To create a professional engineering drawing suitable for academic and construction reference purposes.

5. PROJECT DESCRIPTION AND DESIGN DATA

The below dimensions and material specifications are used for preparing the footing design and drafting drawings in AutoCAD, ensuring proper load transfer, stability, and compliance with standard engineering practices.

Item	Specification
Foundation Type	RCC Stepped Footing
Column Size	300 mm × 300 mm
Top Footing	1000 mm × 1000 mm × 300 mm
Middle Footing	1500 mm × 1500 mm × 300 mm
Bottom Footing	2000 mm × 2000 mm × 300 mm
PCC Thickness	100 mm
Total Footing Depth	1000 mm
Concrete Grade	M25
Steel Grade	Fe500
Foundation Type	Isolated Stepped Footing

6. MAJOR COMPONENTS OF THE STEPPED FOOTING

Component	Function
Column	Transfers structural load to the footing.
Pedestal	Connects the column and footing, distributing loads uniformly.
Top Footing Step	Receives load directly from the column and spreads it over a larger area.
Middle Footing Step	Further distributes the load to reduce soil pressure.
Bottom Footing Step	Provides maximum bearing area and transfers load to the soil.
PCC Bed	Provides a level and clean base for footing construction.
Reinforcement Bars	Increase the strength and durability of the footing.
Concrete Cover	Protects reinforcement from corrosion and environmental effects.
Foundation Soil	Supports the footing and safely carries structural loads.
Footing Edges	Maintain stability and proper load distribution.

7. DRAWING SET AND DETAILS

7.1 Drawing 1 – Stepped Footing Plan

The plan view represents the stepped footing as viewed from above. It shows the column position at the center along with the three footing steps. Dimension lines are provided to indicate the sizes of the column and footing levels clearly.

- Column size: **300 × 300 mm**
- Top footing: **1000 × 1000 × 300 mm**
- Middle footing: **1500 × 1500 × 300 mm**
- Bottom footing: **2000 × 2000 × 300 mm**
- PCC bed: **100 mm thick**
- Centre lines provided for alignment and layout

7.2 Drawing 2 – Front Elevation

The front elevation illustrates the vertical arrangement of the stepped footing below the RCC column. It shows the column, footing steps, PCC bed, and foundation depth. The drawing helps visualize the load transfer path from the structure to the soil.

- Column size: **300 × 300 mm**
- Top footing thickness: **300 mm**
- Middle footing thickness: **300 mm**
- Bottom footing thickness: **300 mm**
- PCC thickness: **100 mm**
- Total footing depth: **1000 mm**
- Number of footing steps: **3**

7.3 Drawing 3 – Section A–A

Section A–A provides a detailed cross-sectional view of the stepped footing. It shows the footing layers, reinforcement arrangement, column connection, footing dimensions, and PCC bed. This drawing is useful for understanding the construction details and load distribution of the foundation system.

- RCC column position
- Reinforcement details
- Three stepped footing levels

- PCC bed below footing
- Soil bearing surface
- Complete vertical dimensions and section details shown clearly in AutoCAD drawings.

8. REINFORCEMENT DETAILS

Member	Thickness / Size	Reinforcement
Column	300 × 300 mm	8 Nos. 16 mm Ø main bars, 8 mm Ø stirrups @ 150 mm c/c
Top Footing	1000 × 1000 × 300 mm	12 mm Ø @ 150 mm c/c, both ways
Middle Footing	1500 × 1500 × 300 mm	12 mm Ø @ 150 mm c/c, both ways
Bottom Footing	2000 × 2000 × 300 mm	12 mm Ø @ 150 mm c/c, both ways
PCC Bed	100 mm thick	Plain Cement Concrete (No reinforcement)

The Reinforcement details shown above are used for the preparation of the **Stepped Footing Design and Drafting using AutoCAD** drawing. The reinforcement arrangement ensures proper load transfer, strength, and durability of the foundation. The final reinforcement details should be verified based on structural design calculations and soil conditions before construction.

9. DIMENSIONS AND TEXT

The stepped footing drawing was prepared in AutoCAD using the specified dimensions and standard drafting practices. The footing consists of three stepped layers of reinforced concrete supporting a central column.

Dimensions Used:

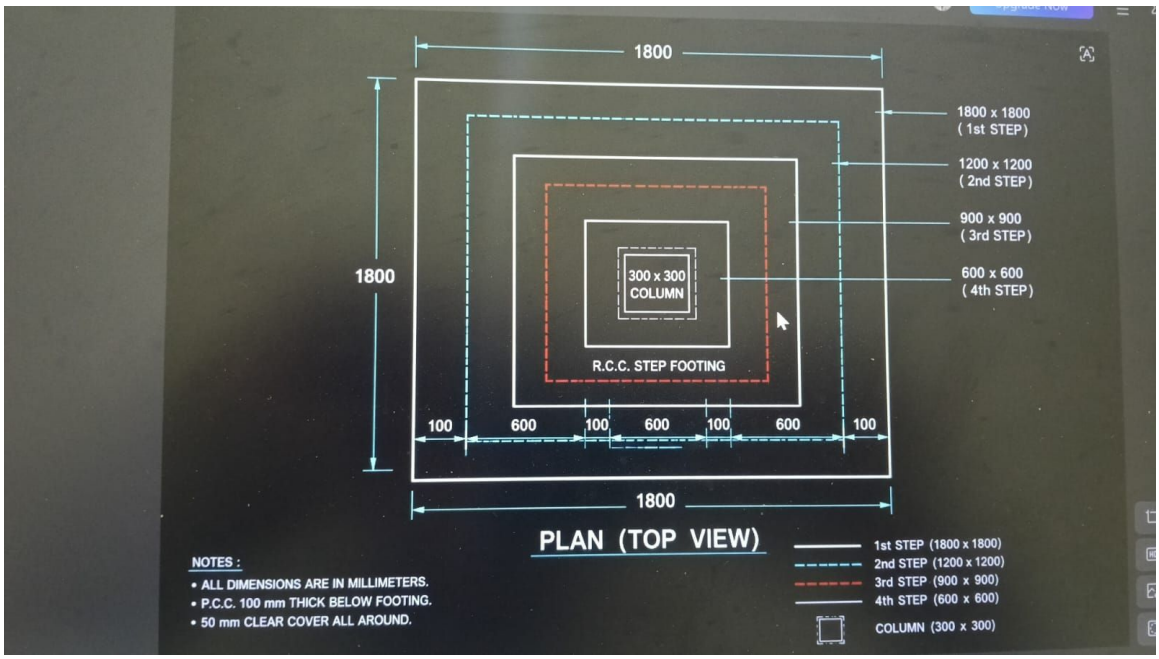
- Column Size: **300 mm × 300 mm**
- Top Footing: **1000 mm × 1000 mm × 300 mm**
- Middle Footing: **1500 mm × 1500 mm × 300 mm**
- Bottom Footing: **2000 mm × 2000 mm × 300 mm**
- Total Footing Depth: **900 mm**
- PCC Bed Thickness: **100 mm**
- Projection of Each Step: **250 mm on all sides**

Drawing Details:

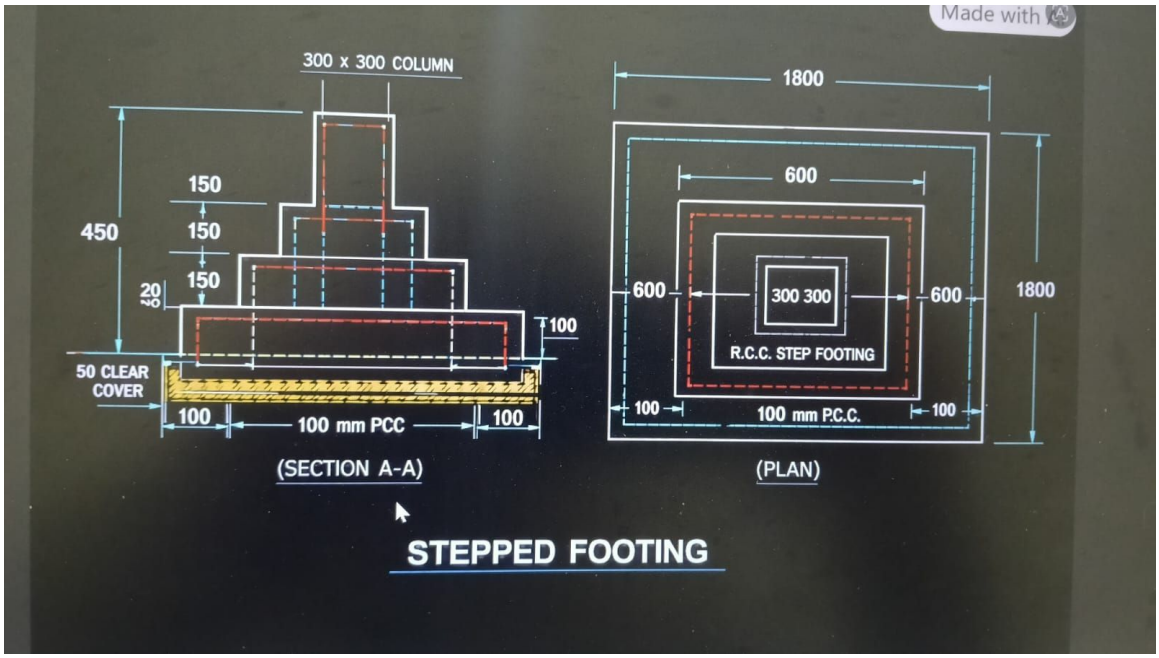
- Plan view showing all footing levels and column position.
- Elevation/section view showing stepped arrangement and depth.
- Dimensions are provided in **millimetres (mm)**.
- Centre lines, dimension lines, and annotations are added for clarity.
- Text height and dimension style are maintained according to standard AutoCAD drafting conventions.

Figure: AutoCAD drawing of RCC Stepped Footing with complete dimensions and labeling.

PLAN(TOP VIEW)



STEPPED FOOTING



10. TECHNOLOGIES AND SOFTWARE USE

SOFTWARE USE

1. **AutoCAD 2023/2024** – For preparing plan, elevation, section, and reinforcement drawings.
2. **Microsoft Word** – For preparing project documentation and reports.
3. **Windows Operating System** – Platform for running AutoCAD and related applications.

TECHNOLOGIES

4. Computer-Aided Design (CAD)
5. Engineering Drawing Standards
6. RCC Foundation Drafting Techniques
7. Reinforcement Detailing
8. Dimensioning and Annotation Methods
9. Layer Management and Hatching Techniques

11. AUTOCAD COMMANDS USED

Category	Commands	Purpose
Drawing	LINE, PLINE, RECTANGLE	Create column and footing outlines
Modification	MOVE, COPY, MIRROR, ROTATE	Position and duplicate objects
Editing	TRIM, EXTEND, FILLET	Clean and modify geometry
Parallel Geometry	OFFSET	Create footing steps and thicknesses
Annotation	TEXT, MTEXT, LEADER	Add notes and labels
Measurement	DIMENSION	Show drawing dimensions
Presentation	HATCH, LAYER	Differentiate materials and drawing elements
Object Editing	STRETCH, SCALE, EXPLODE	Adjust and manage geometry
Repetition	ARRAY	Repeat reinforcement bars
Viewing	ZOOM, PAN	Navigate the drawing efficiently

12. DRAWING TOOLS

Tool	Purpose
LINE	Draws straight lines for footing outlines and construction details.
PLINE (Polyline)	Creates connected line segments as a single object.
RECTANGLE	Draws the column and footing shapes quickly.
CIRCLE	Creates reinforcement symbols and detailing elements.

13. MODIFICATION TOOLS

Tool	Purpose
MOVE	Moves footing components, column, and reinforcement to the required position.
COPY	Creates duplicate footing layers and reinforcement bars.
ROT	Rotates drawing elements to a specified angle.
MIRROR	Creates symmetrical footing and reinforcement details.
OFFSET	Creates parallel footing steps and wall thicknesses.
TRIM	Removes unwanted portions of lines and objects.
EXTEND	Extends objects to meet other boundaries.
STRET	Modifies footing dimensions without redrawing.
SCALE	Changes the size of drawing objects proportionally.

14. APPLICATIONS

- Residential buildings and individual houses.
- Apartment and multi-store building foundations.
- Commercial buildings and office complexes.
- Industrial structures and factory buildings.
- RCC framed structures with isolated columns.
- Structures constructed on soils with moderate bearing capacity.
- Warehouses and storage buildings.
- Educational institutions and public buildings.
- Small bridges and utility structures.
- Foundation systems requiring safe and uniform load distribution.

15. ADVANTAGES OF AUTOCAD-BASED STEPPED FOOTING DRAFTING

- High dimensional accuracy and precision in footing drawings.
- Easy modification of footing sizes, reinforcement, and dimensions.
- Clear representation of plan, elevation, and sectional views.
- Faster preparation and editing of engineering drawings.
- Efficient detailing of reinforcement and annotations.

- Layer-based organization improves drawing management and readability.
- Digital drawings can be stored, shared, and revised easily.
- Reduces drafting errors and improves overall productivity.
- Produces professional and standardized construction drawings.
- Enhances understanding of foundation layout and load distribution.

16. RESULT

A coordinated AutoCAD drawing package for an **RCC Stepped Footing** was successfully prepared using the specified dimensions and detailing information. The completed drawing set consists of the **Footing Plan, Front Elevation, and Section A–A**, supported by dimensions, reinforcement details, footing sizes, column details, foundation information, and general notes. The drawings clearly illustrate the load transfer mechanism from the column to the supporting soil and demonstrate the practical application of AutoCAD in foundation drafting.

17. CONCLUSION

The project demonstrates the practical application of **AutoCAD in civil engineering drafting** through the preparation of an **RCC stepped footing**. The exercise develops an understanding of plan preparation, sectional representation, reinforcement detailing, dimensioning, and drawing management.

The stepped footing arrangement includes a **300 × 300 mm RCC column**, top footing **1000 × 1000 × 300 mm**, middle footing **1500 × 1500 × 300 mm**, bottom footing **2000 × 2000 × 300 mm**, and a **100 mm PCC bed**. The drawings incorporate essential foundation components, including footing steps, reinforcement details, column connections, and construction dimensions.

Overall, the project demonstrates how AutoCAD can convert engineering design information into **accurate, readable, and professional foundation drawings**. The knowledge gained from this project is useful for civil engineering drafting, structural detailing, foundation design, and preparation of technical documentation.

THANK YOU